

Optimal Bidding Strategy

IMC Prosperity – Celestial Gardeners' Guild

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1 Problem Setup

Counterparty reserve prices are uniformly distributed:

$$R \in \{670, 675, 680, \dots, 920\}$$

Total number of possible values:

$$N = \frac{920 - 670}{5} + 1 = 51$$

You submit two bids:

$$b_1 \quad (\text{first bid}), \quad b_2 \quad (\text{second bid}), \quad b_2 > b_1$$

You can resell at fixed price:

$$P = 920$$

2 Trade Mechanics

Define:

$$N_1 = \#\{R : R < b_1\}, \quad N_2 = \#\{R : R < b_2\}$$

Then:

- Trades at b_1 : N_1
- Trades at b_2 : $N_2 - N_1$

Since prices increase in steps of 5:

$$N_1 = \frac{b_1 - 670}{5}, \quad N_2 = \frac{b_2 - 670}{5}$$

$$N_2 - N_1 = \frac{b_2 - b_1}{5}$$

3 Penalty Function

Let $\bar{b}_2$ be the average second bid of all players.

$$q = \begin{cases} 1, & b_2 \geq \bar{b}_2 \\ \left(\frac{920 - \bar{b}_2}{920 - b_2} \right)^3, & b_2 < \bar{b}_2 \end{cases}$$

4 Expected Profit

$$\mathbb{E}[\text{PnL}] = \frac{N_1(920 - b_1) + (N_2 - N_1)(920 - b_2)q}{51}$$

Substitute:

$$\mathbb{E}[\text{PnL}] = \frac{1}{51} \left[\frac{b_1 - 670}{5} (920 - b_1) + \frac{b_2 - b_1}{5} (920 - b_2) q \right]$$

5 Unconstrained Optimum (No Penalty)

Assume:

$$q = 1$$

Maximise:

$$f(b_1, b_2) = \frac{b_1 - 670}{5}(920 - b_1) + \frac{b_2 - b_1}{5}(920 - b_2)$$

Solution:

$$b_1^* = 751, \quad b_2^* = 836$$

6 Expected Profit at Optimum

$$N_1 \approx 17, \quad N_2 - N_1 \approx 17$$

$$\begin{aligned} \mathbb{E}[\text{PnL}] &= \frac{17(920 - 751) + 17(920 - 836)}{51} \\ &= \frac{17(169) + 17(84)}{51} = \frac{4301}{51} \approx 84.33 \end{aligned}$$

7 Risk Adjustment

If:

$$b_2 < \bar{b}_2$$

then:

$$q = \left(\frac{920 - \bar{b}_2}{920 - b_2} \right)^3$$

This introduces a sharp downside risk.

8 Robust Strategy

Choose higher second bid:

$$b_1 = 761, \quad b_2 = 856$$

$$N_1 \approx 19, \quad N_2 - N_1 \approx 19$$

$$\begin{aligned} \mathbb{E}[\text{PnL}] &= \frac{19(920 - 761) + 19(920 - 856)}{51} \\ &= \frac{19(159) + 19(64)}{51} = \frac{4237}{51} \approx 83.08 \end{aligned}$$

9 Final Recommendation

- Risk-neutral optimum:

$$(751, 836)$$

- Robust competition-aware optimum:

$$(761, 856)$$

Reasoning Summary

1. The reserve price is uniformly distributed across 51 equally spaced values from 670 to 920.
2. The first bid captures all reserve prices below b_1 , while the second bid captures the additional reserve prices between b_1 and b_2 .
3. Without penalty, the best pair of bids is approximately (751, 836), giving an expected profit of about 84.33.
4. Because bidding too low on the second bid can trigger a cubic penalty, a slightly higher second bid is safer in competition.
5. A robust practical choice is (761, 856), which gives slightly lower raw expected profit but materially reduces downside risk.

Appendix: Key Formulae Reference

$$N = \frac{920 - 670}{5} + 1 \tag{1}$$

$$N_1 = \frac{b_1 - 670}{5} \tag{2}$$

$$N_2 = \frac{b_2 - 670}{5} \tag{3}$$

$$q = \begin{cases} 1, & b_2 \geq \bar{b}_2 \\ \left(\frac{920 - \bar{b}_2}{920 - b_2} \right)^3, & b_2 < \bar{b}_2 \end{cases}$$

$$\mathbb{E}[\text{PnL}] = \frac{1}{51} \left[\frac{b_1 - 670}{5} (920 - b_1) + \frac{b_2 - b_1}{5} (920 - b_2) q \right]$$